

RAG Pipeline Visualizer — learn every part

A document enters on the left and leaves as a grounded, cited answer on the right.
This sheet maps the whole lab: the 14-stage signal path, every instrument you can open, and the order to learn them in. Nothing is a black box — every stage keeps its real artifacts, and you can open, tune, break, and replay all of them.

14

PIPELINE STAGES

5

PERSONAS

8

GUIDED CHAPTERS

768-dim

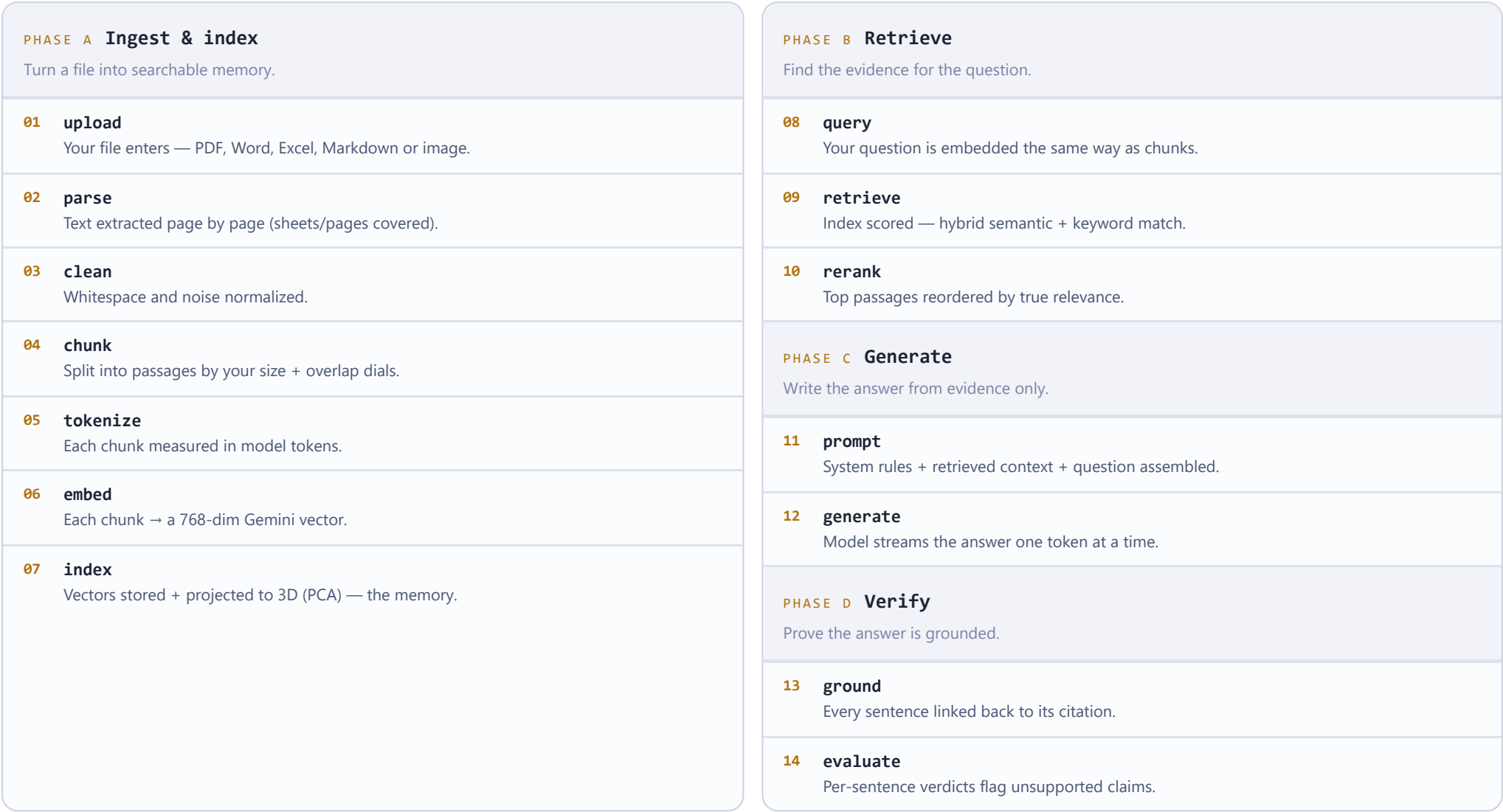
REAL VECTORS

≤ 10 MB

PDF · WORD · EXCEL · MD · IMAGE

1 The signal path — 14 stages, left to right

click any node → Inspector opens its real data



3 The instruments – every panel, and how to use it

what it shows · one thing to try

RIGHT PANEL

Inspector

The X-ray of any stage. Click a node to see its real artifacts: extracted pages, chunks, token counts, a live vector sample, retrieval scores, the assembled prompt.

► Open the **embed** node and read the first 14 of a chunk's 768 dimensions.

3D VIEW

Embedding Universe

Every chunk plotted in space by meaning — a PCA projection of the real 768-dim vectors, with clusters labeled. Similar passages sit close together.

► Ask a question and watch which cluster the retrieved chunks come from.

SCRUBBER

Timeline

The recorded run as a scrubbable strip. Step backward and forward through every stage to see exactly when each artifact appeared.

► Drag the scrubber back to **chunk** and replay the build forward.

DOCK · PARAMETERS

Parameters

The dials that decide quality: chunk size & overlap, top-K, threshold, hybrid weighting, rerank, temperature, token budgets, and the system prompt itself.

► Move one dial, re-ask, and watch retrieval and the answer change. (See §5.)

DOCK · METRICS

Metrics

How the run performed: a cost meter (tokens → \$ at paid scale), a latency distribution across stages, and an evaluation radar of answer quality.

► Compare latency before and after turning **rerank** off.

DOCK · A/B

A/B Playground

Run the same question under two configurations, side by side, and get a plain **verdict** on which config won — opinion ends where measurement begins.

► Set topK 2 vs 6 and let the eval scores settle the argument.

DOCK · AI LAB

AI Lab

Hypothesis-first experiments: predict the outcome, then run a **sabotage preset** to break retrieval on purpose and diagnose the damage.

► Predict, then run a sabotage preset and find the failure in the metrics.

ASSISTANT

Coach

Contextual insights that appear as you work — pointing out what just changed and why it matters, tuned to your persona.

► Change a parameter and read the coach note that follows.

OVERLAY · TRACE

Detective

A grounded answer is a checkable answer. Pick one sentence and follow its citation backward to the exact chunk — and page — it came from.

► Click a cited sentence and trace it to its source passage.

OVERLAY · THEATER

Brain / Generation Theater

Watch the model think in four acts: the briefing arrives, the working set is fixed, tokens stream out, and citation markers appear live as it writes.

► Open it and watch which chunk earns its citation, and when.

OVERLAY · HEATMAP

Chunk Profile

A per-chunk close-up: size, tokens, overlap and how strongly it scored — so you can see which passages carry the answer.

► Open the top-scoring chunk and see why it was retrieved.

OVERLAY · PROMPT

Prompt MRI

Dissect the exact prompt sent to the model — system rules, retrieved context, and question — with the token budget each block consumed.

► Shrink the context budget and watch which evidence gets dropped.

4 The learning path – do these in order

the built-in 8-chapter journey

1 Ingest a document

Watch a file become clean text → chunks → vectors in seconds.

2 Ask your first question

Your question becomes a vector; the index finds the evidence; the model answers from it.

3 Open a pipeline node

Prove nothing is a black box — every stage shows its real artifacts.

4 Trace an answer to its source

Follow one sentence back through its citation to the exact passage and page.

5 Tune a parameter & re-ask

Move one dial, ask again, watch retrieval, prompt and answer respond.

6 Break it on purpose

Run a sabotage preset and diagnose the damage in the metrics.

7 Compare two configurations

Run the same question two ways; let the evaluation scores decide.

8 Present it to someone

The final test of understanding: narrate the pipeline to a friend or a room.

DIAL	DEFAULT	WHAT IT CONTROLS	TURN IT UP →
chunk size	600	Characters per passage before splitting.	Bigger, fewer chunks — more context each, but coarser retrieval.
chunk overlap	80	Characters shared between neighbouring chunks.	Fewer split sentences at boundaries, at some redundancy cost.
top-K	4	How many chunks are retrieved per question.	More evidence in the prompt — richer, but noisier and pricier.
threshold	0.25	Minimum similarity a chunk needs to qualify.	Stricter — only close matches survive; may return nothing.
hybrid α	0.70	Blend of semantic (meaning) vs keyword match.	Leans on meaning over exact words.
rerank	on	Reorders retrieved chunks by true relevance.	Better ordering; a little more latency.
temperature	1.0	Randomness of the generated answer.	More varied wording; higher drift from the evidence.
max tokens	600	Length cap on the answer.	Room for longer answers.
context budget	2000	Token ceiling for retrieved context in the prompt.	More evidence fits; lower-ranked chunks stop getting cut.
system prompt	edit	The rules the model must obey (cite, refuse if absent).	Rewrite it to change how strictly it grounds.

How to read this sheet: follow §1 to see the flow, drive from §2, open the instruments in §3, then learn in the §4 order while tuning the §5 dials.
Every model call runs on Google Gemini (embeddings + generation + vision), with an optional OpenAI fallback.

RAG Pipeline Visualizer · [/learn/rag](#)